Verasight Data Scientist Case Study

Attached, you will find four files:

1. A sample of registered Verasight panelists (users.rds), who take repeated surveys after creating their accounts
2. A dataset with responses from one specific recent survey (2024-054_responses.rds)
3. A reference file with information about the survey questions in the response dataset (2024-054_reference.rds)
4. An aggregate response dataset with survey responses from many 2024 Verasight projects (full-response-db.rds)

Using these files, complete the following tasks in R and comment the code appropriately. When finished, please send all code and output to careers@verasight.io. **Please do not spend more than 3 hours on this case study.** If you don't know how to do a certain step of the case study, please outline the questions that you would ask our team to be able to complete it. If you run out of time, please briefly describe the steps you would take to complete the remaining tasks.

You may use generative AI to complete these tasks. If you do, please submit an ai_memo.md file that describes how you used AI tools to complete the task. Your memo should include:

- Which model(s) you used and why
- Methods you used to communicate with LLMs (e.g., browser, CLI, IDE extension)
- Any harnessing around LLMs (e.g., custom agents, skills)
- Representative examples of LLM prompts
- Methods you use to check and ensure accuracy of LLM output
- LLM output that you corrected, disagreed with, or had to steer toward a better solution

Task 1

Construct and run a tabulation function

Create a single, documented, reusable R function (plus any small helpers it needs) to produce weighted proportion tables like the following for each survey outcome by each demographic variable in the recent response dataset (2024-054_responses.rds), with the question wording attached.

These data are from a nationally representative survey of U.S. adults.

Your function should be constructed such that a teammate could understand and use your function on a *different* Verasight survey without editing its internals.

Response	18-29	30-49	50-64	65+
Always	28%	32%	24%	23%
Sometimes	56%	50%	51%	52%
Seldom	12%	14%	17%	18%
Never	5%	4%	8%	7%
Don't Know	0%	0%	0%	0%

Task 2

When analyzing data from our survey respondent community, we are sometimes interested in a concept we refer to as "density." This concept helps us measure whether our survey responses are concentrated among a small number of panelists, or spread out more evenly over a larger number of users/panelists.

1. Using the aggregate response database (response-db.rds), please answer the following question: Out of all Verasight users in the database, what is the smallest percent of users who can account for 50% of all recorded survey responses?
2. Calculate the same metric, but restricted to Verasight users who registered within the last 90 days.
3. Create a visualization showing the density in more detail – in other words, visualize the percent of surveys accounted for at various percentiles of response contribution. (For example, what is the minimum percent of users who account for 10%/20%/30% of survey responses?)

Task 3

Open-ended (or "free response") survey items are rich sources of information about respondents' thoughts and feelings, but also about the quality of the data. Consider the following questions that may be included in Verasight surveys:

- "In your own words, what is the most important issue facing the United States today and why?"
- "What qualities do you think a good leader should have?"
- "Please describe any issues or problems you faced in completing this survey today."

Reading through open-ended responses to detect potentially fraudulent survey behavior can be a time-intensive manual process. Imagine you are tasked with using AI tools to build a fully or partially automated workflow for the Data Science team at Verasight to assess responses to questions like these, across many different surveys, to identify potentially fraudulent survey behavior. What tools would you use? What would your initial design or setup look like? How might you iterate on the first version of your process to improve its accuracy and validity?

Task 4

Included below are descriptions of 4 types of projects you could work on as a Data Scientist at Verasight. Read through the descriptions and then answer the questions below.

1. A nationally-representative survey of 3,000 US adults fielded from the Verasight panel. Questions were submitted by conference attendees. In addition to the standard deliverables of raw data, codebook, and survey weights, this project also requires a public-facing topline & crosstab report. Fielding & deliverable preparation should take 2 weeks.
2. A survey of 1,500 voters in Wyoming sampled from voter file records. The sales team notes that the client may have demographic quotas. Fielding should take 3 weeks.
3. A survey of 300 software developers who program with AI models. The client would like to be able to weight the survey results to the population of eligible software developers. The timeline is 6 weeks to prepare and 4 weeks to field.
4. A longitudinal survey of 1,000 working women. There are 4 waves, each a month apart. Fielding should start tomorrow.

Part 1:

For each project, identify the 1-3 of the most important questions you would ask before beginning work. Your questions should help you clarify the project scope, the goals of the client, assess feasibility, and understand the specific data needs of the project.

Questions may cover (but are not limited to) sampling frame creation, weighting or analysis requirements, deliverables, timelines, or client expectations.

Assume you have the resources and training you need to complete individual tasks successfully - ask questions to help understand the *scope and content* of the projects, not on the mechanics of how to execute the tasks themselves.

Part 2:

Now assume you receive answers to the questions you included above. Given what you know now, please rank these projects from 1-4 in order of how time intensive you think they will be for you as a Data Scientist. The project you rank first should be the project you expect to take the most of your time, and the project you rank 4th should be the project you expect to take the least amount of your time. Be explicit about the assumptions you make and provide reasoning for the ranking. (Note: There are no right or wrong answers for this question - we want to get insight into your thought process.)